6.1.10 Examples **(a)** The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := x^5 + 4x + 3$ is continuous and strictly monotone increasing (since it is the sum of two strictly increasing functions). Moreover, $f'(x) = 5x^4 + 4$ is never zero. Therefore, by Theorem 6.1.8, the inverse function $g = f^{-1}$ is differentiable at every point. If we take $c = 1$, then since $f(1) = 8$, we obtain $g'(8) = g'(f(1)) = 1/f'(1) = 1/9$.

(b) Let $n \in \mathbb{N}$ be even, let $I := [0, \infty)$, and let $f(x) := x^n$ for $x \in I$. It was seen at the end of Section 5.6 that f is strictly increasing and continuous on I, so that its inverse function $g(y) := y^{1/n}$ for $y \in J := [0, \infty)$ is also strictly increasing and continuous on J. Moreover, we have $f'(x) = nx^{n-1}$ for all $x \in I$. Hence it follows that if $y > 0$, then $g'(y)$ exists and

$$g'(y) = \frac{1}{f'(g(y))} = \frac{1}{n(g(y))^{n-1}} = \frac{1}{ny^{(n-1)/n}}.$$

Hence we deduce that

$$g'(y) = \frac{1}{n} y^{(1/n)-1} \quad \text{for} \quad y > 0.$$

However, g is *not* differentiable at 0. (For a graph of f and g, see Figures 5.6.4 and 5.6.5.)

(c) Let $n \in \mathbb{N}$, $n \neq 1$, be odd, let $F(x) := x^n$ *for* $x \in \mathbb{R}$, and let $G(y) := y^{1/n}$ be its inverse function defined for all $y \in \mathbb{R}$. As in part (b) we find that G is differentiable for $y \neq 0$ and that $G'(y) = (1/n)y^{(1/n)-1}$ for $y \neq 0$. However, G is not differentiable at 0, even though G is differentiable for all $y \neq 0$. (For a graph of F and G, see Figures 5.6.6 and 5.6.7.)

(d) Let $r := m/n$ be a positive rational number, let $I := [0, \infty)$, and let $R(x) := x^r$ for $x \in I$. (Recall Definition 5.6.6.) Then R is the composition of the functions $f(x) := x^m$ and $g(x) := x^{1/n}$, $x \in I$. That is, $R(x) = f(g(x))$ for $x \in I$. If we apply the Chain Rule 6.1.6 and the results of (b) [or (c), depending on whether n is even or odd], then we obtain

$$\begin{aligned} R'(x) &= f'(g(x))g'(x) = m\left(x^{1/n}\right)^{m-1} \cdot \frac{1}{n} x^{(1/n)-1} \\ &= \frac{m}{n} x^{(m/n)-1} = rx^{r-1} \end{aligned}$$

for all $x > 0$. If $r > 1$, then it is an exercise to show that the derivative also exists at $x = 0$ and $R'(0) = 0$. (For a graph of R see Figure 5.6.8.)

(e) The sine function is strictly increasing on the interval $I := [-\pi/2, \pi/2]$; therefore its inverse function, which we will denote by Arcsin, exists on $J := [-1, 1]$. That is, if $x \in [-\pi/2, \pi/2]$ and $y \in [-1, 1]$ then $y = \sin x$ if and only if $\text{Arcsin } y = x$. It was asserted (without proof) in Example 6.1.7(d) that sin is differentiable on I and that $D \sin x = \cos x$ for $x \in I$. Since $\cos x \neq 0$ for x in $(-\pi/2, \pi/2)$ it follows from Theorem 6.1.8 that

$$\begin{aligned} D \operatorname{Arcsin} y &= \frac{1}{D \sin x} = \frac{1}{\cos x} \\ &= \frac{1}{\sqrt{1 - (\sin x)^2}} = \frac{1}{\sqrt{1 - y^2}} \end{aligned}$$

for all $y \in (-1, 1)$. The derivative of Arcsin does *not* exist at the points -1 and 1. □

Exercises for Section 6.1

1. Use the definition to find the derivative of each of the following functions:
   (a) $f(x) := x^3$ for $x \in \mathbb{R}$,
   (b) $g(x) := 1/x$ for $x \in \mathbb{R}$, $x \neq 0$,
   (c) $h(x) := \sqrt{x}$ for $x > 0$,
   (d) $k(x) := 1/\sqrt{x}$ for $x > 0$.

2. Show that $f(x) := x^{1/3}$, $x \in \mathbb{R}$, is not differentiable at $x = 0$.
3. Prove Theorem 6.1.3(a), (b).
4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := x^2$ for x rational, $f(x) := 0$ for x irrational. Show that f is differentiable at $x = 0$, and find $f'(0)$.
5. Differentiate and simplify:
   (a) $f(x) := \dfrac{x}{1+x^2}$, (b) $g(x) := \sqrt{5-2x+x^2}$,
   (c) $h(x) := (\sin x^k)^m$ for $m, k \in \mathbb{N}$, (d) $k(x) := \tan(x^2)$ for $|x| < \sqrt{\pi/2}$.
6. Let $n \in \mathbb{N}$ and let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := x^n$ for $x \geq 0$ and $f(x) := 0$ for $x < 0$. For which values of n is f' continuous at 0? For which values of n is f' differentiable at 0?
7. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable at c and that $f(c) = 0$. Show that $g(x) := |f(x)|$ is differentiable at c if and only if $f'(c) = 0$.
8. Determine where each of the following functions from $\mathbb{R}$ to $\mathbb{R}$ is differentiable and find the derivative:
   (a) $f(x) := |x| + |x+1|$, (b) $g(x) := 2x + |x|$,
   (c) $h(x) := x|x|$, (d) $k(x) := |\sin x|$.
9. Prove that if $f : \mathbb{R} \to \mathbb{R}$ is an **even function** [that is, $f(-x) = f(x)$ for all $x \in \mathbb{R}$] and has a derivative at every point, then the derivative f' is an **odd function** [that is, $f'(-x) = -f'(x)$ for all $x \in \mathbb{R}$]. Also prove that if $g : \mathbb{R} \to \mathbb{R}$ is a differentiable odd function, then g' is an even function.
10. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) := x^2 \sin(1/x^2)$ for $x \neq 0$, and $g(0) := 0$. Show that g is differentiable for all $x \in \mathbb{R}$. Also show that the derivative g' is not bounded on the interval $[-1, 1]$.
11. Assume that there exists a function $L : (0, \infty) \to \mathbb{R}$ such that $L'(x) = 1/x$ for $x > 0$. Calculate the derivatives of the following functions:
   (a) $f(x) := L(2x+3)$ for $x > 0$, (b) $g(x) := (L(x^2))^3$ for $x > 0$,
   (c) $h(x) := L(ax)$ for $a > 0$, $x > 0$, (d) $k(x) := L(L(x))$ when $L(x) > 0$, $x > 0$.
12. If $r > 0$ is a rational number, let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := x^r \sin(1/x)$ for $x \neq 0$, and $f(0) := 0$. Determine those values of r for which $f'(0)$ exists.
13. If $f : \mathbb{R} \to \mathbb{R}$ is differentiable at $c \in \mathbb{R}$, show that
$$f'(c) = \lim(n\{f(c + 1/n) - f(c)\}).$$
However, show by example that the existence of the limit of this sequence does not imply the existence of $f'(c)$.
14. Given that the function $h(x) := x^3 + 2x + 1$ for $x \in \mathbb{R}$ has an inverse h^{-1} on $\mathbb{R}$, find the value of $(h^{-1})'(y)$ at the points corresponding to $x = 0, 1, -1$.
15. Given that the restriction of the cosine function cos to $I := [0, \pi]$ is strictly decreasing and that $\cos 0 = 1$, $\cos \pi = -1$, let $J := [-1, 1]$, and let Arccos: $J \to \mathbb{R}$ be the function inverse to the restriction of cos to I. Show that Arccos is differentiable on $(-1, 1)$ and $D \operatorname{Arccos} y = (-1)/(1-y^2)^{1/2}$ for $y \in (-1, 1)$. Show that Arccos is not differentiable at -1 and 1.
16. Given that the restriction of the tangent function tan to $I := (-\pi/2, \pi/2)$ is strictly increasing and that $\tan(I) = \mathbb{R}$, let Arctan: $\mathbb{R} \to \mathbb{R}$ be the function inverse to the restriction of tan to I. Show that Arctan is differentiable on $\mathbb{R}$ and that $D \operatorname{Arctan}(y) = (1+y^2)^{-1}$ for $y \in \mathbb{R}$.
17. Let $f : I \to \mathbb{R}$ be differentiable at $c \in I$. Establish the **Straddle Lemma:** Given $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that if $u, v \in I$ satisfy $c - \delta(\varepsilon) < u \leq c \leq v < c + \delta(\varepsilon)$, then we have $|f(v) - f(u) - (v-u)f'(c)| \leq \varepsilon(v-u)$. [*Hint*: The $\delta(\varepsilon)$ is given by Definition 6.1.1. Subtract and add the term $f(c) - cf'(c)$ on the left side and use the Triangle Inequality.]